
Herd Immunity and Learning Externalities in Markets of Adaptive Models

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Abstract

1 A learning agent can satisfy every single-agent stability criterion and still help
2 destabilize the market it operates in. When adaptive agents share an environment
3 that reacts to their decisions, each agent’s retraining reshapes the data every other
4 agent learns from next, a *learning externality* that no agent internalizes. Its strength
5 is governed not by the number of firms but by the *effective number of independent*
6 *learners*, the leading eigenvalue of the correlation matrix of firms’ response Jaco-
7 bians, so foundation-model concentration enters the stability condition as a term
8 rather than as a concern: fifty firms fine-tuning one vendor’s model are dynami-
9 cally close to one large learner. Three closed-form levers follow from a reduction
10 lemma we derive in the main text. A crowding-cadence frontier bounds how often
11 a market can retrain and closes past a critical crowding level. A herd-immunity
12 threshold makes the market stable exactly when the corrected fraction exceeds
13 $(1 - 1/m_N)/e$ at correction efficacy e , the imperfect-vaccine coverage law with
14 the systemic modulus as the reproduction number. A Pigouvian wedge prices the
15 externality, and every market of two or more firms over-adapts. The last two levers
16 are substitutes along a frontier we compute. In a shared-pool order-flow market
17 the mechanism reproduces a published $1.74\times$ and $3.16\times$ amplification at two and
18 three firms, bit for bit.

19 1 Introduction

20 A learning agent can pass every stability test we know how to run and still help destabilize the market
21 it operates in.

22 The test we know how to run is a single-agent test. A model is evaluated, stress-tested and certified
23 alone, against a distribution treated as given. When the deployed model reshapes that distribution, the
24 performative prediction literature supplies the correction: repeated retraining converges if and only if
25 a modulus $m = \varepsilon\beta/\gamma$ stays below one, where ε measures how strongly the environment responds
26 and γ is the learner’s own objective curvature (Perdomo et al., 2020). Certify $m < 1$ and the loop is
27 safe.

28 It is not, once the agent has company. Consider dealers whose quotes reshape a common pool
29 of informed flow, recommenders competing for one attention pool, or lenders pricing against one
30 population of borrowers. Each agent’s retraining changes the data every *other* agent will learn from
31 next, and no agent internalizes that effect. We call this a **learning externality**: every participant
32 behaves rationally, every participant passes its own test, and the interaction amplifies a loop none
33 of them chose. The intuition is not new. Banks that each diversify optimally hold similar portfolios
34 and fail together (Beale et al., 2011; Wagner, 2010), and funds running similar strategies unwound
35 simultaneously in August 2007 with the crowding invisible from any single risk report (Khandani

and Lo, 2007). What is new is that the agents now *learn*, which changes both what determines the danger and what can be done about it.

What determines it is not how many firms there are. Let firm i have a response Jacobian E_i describing how its own deployment reshapes the flow it faces, and let R be the correlation matrix of the vectorized E_i . We show the market’s stability is governed by $m_N = N_{\text{eff}} m_1$ with $N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1)$, where κ is the spillover between firms and $\lambda_{\max}(R)$ is the **effective number of independent learners**: a hundred firms perturbing the environment in a hundred orthogonal directions are dynamically one firm. With s the fraction of each response attributable to a common model, vendor or pretraining corpus, this reads $m_N = m_1(1 + \kappa s(N - 1))$, so **the effective number of learners is the number of independent models, not the number of firms**. The quantity a competition regulator measures is the wrong one: fifty equal-share firms have a minimal Herfindahl index and look perfectly competitive, and if all fifty share one vendor they are dynamically a monoculture (Hirschman, 1964).

What can be done about it takes three forms, and two are substitutes. Because every result is stated in m_N , each inherits the supply chain for free. *Slow down*: K -step retraining faces a window $K < K_{\max}(m_N)$ that narrows as crowding rises and closes past a critical crowding level beyond which no frequency is safe. *Correct*: feedback-aware updates (Izzo et al., 2021) run inside the market produce exactly the epidemiological herd-immunity law with the systemic modulus as the reproduction number. *Diversify*: that threshold rises in s , so a market reaches stability along either axis, and the iso-stability frontier between the two is this paper’s most directly usable object. Pricing the externality completes the policy triple, and the decentralized equilibrium over-adapts for every $N \geq 2$.

Contributions. (1) The effective number of independent learners as the reproduction number of a market of adaptive models, derived from a reduction lemma, turning foundation-model concentration into a term in a stability condition, with two counterexamples invalidating mean-similarity diversity indices. (2) The crowding-cadence frontier and its critical crowding level, composing lazy deployment with multi-agent amplification. (3) A mixed-market herd-immunity theorem, exact at finite correction strength, whose clean form is the imperfect-vaccine coverage law and which carries a critical efficacy. (4) The substitution frontier between diversity and corrected learning, which exists only once (1) and (3) are expressed in one quantity. (5) The Pigouvian wedge and the over-adaptation corollary.

2 Related work

Monoculture, and why it is static. Kleinberg and Raghavan (2021) show that when many firms adopt the same algorithm welfare can fall even when the shared algorithm is the better one, because correlated selection destroys the option value of independent evaluation; Bommasani et al. (2022) formalize systemic exclusion when many deployments inherit one model. Both are cross-sectional, measured at one point in time on one cohort, so neither can express a stability boundary or a rate. A market can be statically fine and dynamically fragile at the same shared-model fraction.

Multiplayer performative prediction, and the containment relation. Narang et al. (2023) is the nearest dynamics: multiple learners in a shared decision-dependent environment, with equilibrium concepts and convergence conditions for repeated retraining. The containment relation is worth stating, since Theorem 2 should not have to be taken on trust as more than a corollary. **Theirs bounds scalars**: the sensitivity of the joint distribution map to the players’ decisions, measured by Lipschitz constants, against their objective curvature, uniformly over the joint action. **Ours bounds a spectral radius**, $m_N = m_1[1 + \kappa(\lambda_{\max}(R) - 1)]$, in which those scalars appear only inside m_1 . **Ours is strictly sharper on the class their constants describe, and is not implied by theirs**. Sharper, because at fixed constants N_{eff} still ranges over $[1, 1 + \kappa(N - 1)]$, so a uniform bound must hold at the worst alignment structure while ours need not, and every market between the two is one their condition rejects and ours admits. Not implied, because nothing in their framework represents *which direction* each agent perturbs the environment: two markets with identical constants and opposite alignment structure are indistinguishable to them and differ in stability for us. The substitution $\mathbf{1}\mathbf{1}^\top \rightarrow R$ adds an object rather than tightening a constant, and that object, not the dynamics, carries monoculture, provenance, the supply chain and every policy statement below. The single-learner

88 lazy-deployment slope of Mendler-Dünner et al. (2020) is what Theorem 3 composes with multi-agent
 89 amplification, a composition not previously made; the corrected update of Izzo et al. (2021) is what
 90 Theorem 4 puts into the game, having only ever been run single-learner.

91 **Systemic risk, epidemiology, and the statistic.** Beale et al. (2011) asked this question of banks,
 92 showing individually prudent portfolio choices can be collectively destabilizing; their agents do not
 93 learn, so the homogeneity is a portfolio overlap rather than an alignment of feedback directions,
 94 and there is no second instrument to trade diversity against. Diekmann et al. (1990) converts the
 95 epidemiological parallel from analogy into identity: R_0 in a heterogeneous population is *defined* as
 96 the spectral radius of a next-generation operator, and m_N is the spectral radius of a linearized joint
 97 retraining operator, so the two thresholds are the same statement and the vaccination law transfers
 98 with its refinements. The largest eigenvalue of a correlation matrix as an effective count is meanwhile
 99 standard in random-matrix finance (Laloux et al., 1999; Plerou et al., 2002), signal processing (Roy
 100 and Vetterli, 2007) and ecology (Hill, 1973); the novelty is the matrix it is computed from, response
 101 Jacobians rather than returns, and the condition it enters. Work reading macroeconomic state off
 102 public sentiment (Nagarajan and Mittal, 2024) estimates through an exogenous channel, whose
 103 accuracy does not depend on the state of the system being measured, while the estimator of Section 9
 104 reads an endogenous one that sharpens as the market approaches instability, so the two face opposite
 105 sample-complexity problems near the event of interest.

106 3 Setup: private against systemic stability

107 **The base result.** Consider N symmetric dealers quoting the same instruments and sharing one
 108 pool of informed order flow, coupled by a spillover $\kappa \in [0, 1]$: one dealer tightening its quotes
 109 raises the toxic flow routed to every dealer, in proportion κ . Each periodically retrains on the flow
 110 it has observed and redeploys, and a firm in isolation is stable exactly when its own performative
 111 modulus satisfies $m_1 < 1$. The joint retraining map has Jacobian $J = -m_1[(1 - \kappa)I + \kappa \mathbf{1}\mathbf{1}^\top]$,
 112 whose dominant common mode sits at $-m_1 N_{\text{eff}}$ with $N_{\text{eff}} = 1 + \kappa(N - 1)$, so the market crosses
 113 into instability a factor N_{eff} earlier than any individual dealer’s own loop would. Measured in a
 114 shared-pool simulator with real spreads, inventory and liquidity feedback, amplification is $1.74\times$ at
 115 $N = 2$ and $3.16\times$ at $N = 3$ against a linear prediction of 2 and 3 (Nagarajan and Ashok, 2026). This
 116 is inherited scaffolding; everything below replaces one object in it.

117 **Two questions, and they are not the same question.** Is each agent stable alone, $m_1 < 1$? And
 118 is the collection stable when its members interact, $m_N < 1$? A shared environment separates
 119 them, because the loop closes through that environment and nothing in dealer A’s validation set
 120 contains dealer B. **Agent count is not a descriptive characteristic of a market; it enters the**
 121 **stability condition.** Read as economics, this is an externality, and it is *technological rather than*
 122 *pecuniary* (Buchanan and Stubblebine, 1962): firms affecting each other only through prices impose
 123 no market failure, since the price movement is a transfer, while firms that reshape the distribution
 124 their competitors learn from change the production technology of everyone’s forecast, which is a
 125 genuine efficiency loss. We call m_N the **feedback reproduction number**: the factor by which a
 126 common perturbation is amplified per retraining round.

127 **Objects and standing assumptions.** Firm i has a response Jacobian $E_i \in \mathbb{R}^{d \times d}$. The *alignment*
 128 *matrix* $R = (r_{ij})$ collects $r_{ij} = \langle \text{vec } E_i, \text{vec } E_j \rangle / (\|E_i\|_F \|E_j\|_F)$, a correlation matrix of feedback
 129 directions rather than of returns; it is positive semidefinite with unit diagonal.

130 **Assumption 1** (Standing). (A1) All spectral statements linearize the joint retraining map around
 131 the joint equilibrium, with the simulator as the nonlinear check. (A2) Theorem 2 is exact for equal
 132 moduli $m_i = m_1$; unequal moduli give a two-sided bound. (A3) Firms retrain on a common clock
 133 at a common cadence K . (A4) Firms couple through one pool with a scalar spillover κ . No sign
 134 condition on R is assumed; the one place one is needed is stated where it is used, in Section 7.

135 **The reduction.** The substitution of R for $\mathbf{1}\mathbf{1}^\top$ is the whole paper, so it is derived rather than
 136 posited.

137 **Lemma 1** (Reduction). Assume (H1) firm i ’s deviation from equilibrium is a scalar x_i along its unit
 138 response direction $\hat{E}_i = E_i / \|E_i\|_F$, so the state is $x \in \mathbb{R}^N$ rather than $\mathbb{R}^{Nd}$; (H2) $\|E_i\|_F = \varepsilon$ for

Table 1: Anchors of Theorem 2. The simplex is the sharper counterexample to mean-based diversity: it *minimizes* mean alignment yet exceeds orthogonality’s $\lambda_{\max}$.

Configuration	$\lambda_{\max}(R)$	N_{eff}	Reading
Monoculture, $R = \mathbf{1}\mathbf{1}^\top$	N	$1 + \kappa(N - 1)$	the inherited law
Orthogonal, $R = I$	1	1	a hundred firms are dynamically one firm
Simplex, $r_{ij} = -1/(N-1)$	$N/(N-1)$	$1 + \kappa/(N-1)$	maximal diversity, and the mode labels swap
Supply chain, share s	$1 + s(N - 1)$	$1 + \kappa s(N - 1)$	the number of independent <i>models</i>

every i , which is (A2) restated in the new object; (H3) firm i retrains against the component of the pool distortion along $\hat{E}_i$, feeling its own contribution in full and each competitor’s with weight κ ; and (H4) firm i minimizes a γ -strongly-convex objective with sensitivity β to the felt distortion, so $x_i^+ = -(\beta/\gamma) \text{felt}_i$. Then the joint retraining map is linear with Jacobian

$$J = -m_1[(1 - \kappa)I + \kappa R], \quad m_1 = \varepsilon\beta/\gamma. \quad (1)$$

Proof sketch. The pool carries the aggregate distortion $Z = \sum_j x_j E_j$, so by (H3) firm i feels $\text{felt}_i = \langle \hat{E}_i, x_i E_i \rangle + \kappa \sum_{j \neq i} \langle \hat{E}_i, x_j E_j \rangle$. By (H2), $\langle \hat{E}_i, E_j \rangle = \varepsilon r_{ij}$, and since $r_{ii} = 1$ the own term folds into the sum, leaving $\text{felt}_i = \varepsilon[(1 - \kappa)x_i + \kappa(Rx)_i]$. Applying (H4) coordinatewise gives (1), with m_1 the single-firm modulus of Perdomo et al. (2020). $\square$

Alignment enters through (H3): a firm senses the pool through its own response channel, so two firms perturbing the environment in orthogonal directions do not read each other at all, however large those perturbations. (H1) is the substantive hypothesis; without it J carries a Kronecker structure on $\mathbb{R}^{Nd}$ whose radius is not in general $N_{\text{eff}} m_1$, the block-secular problem deferred in Section 9.

4 Result 1: the effective number of independent learners

Counting firms is wrong, and the correction is where monoculture and supply-chain content enter. $\lambda_{\max}(R)$ runs from 1, all responses orthogonal, to N , monoculture.

Theorem 2 (Alignment sets the reproduction number). *Under Assumption 1 with equal moduli and heterogeneous response directions, the joint Jacobian is (1) and the market’s spectral radius is*

$$m_N = N_{\text{eff}} m_1, \quad N_{\text{eff}} = 1 + \kappa(\lambda_{\max}(R) - 1), \quad (2)$$

so the market is stable if and only if $m_N < 1$.

Setting $R = \mathbf{1}\mathbf{1}^\top$ recovers the symmetric multi-dealer law of Nagarajan and Ashok (2026) exactly, so the published result is not generalized away but located: it is the monoculture corner, where every firm’s feedback points the same way. Four anchors follow, each checked independently (Table 1). $N_{\text{eff}} \geq 1$ always, so interaction never stabilizes a market below what its members achieve alone.

The supply chain. Decompose $E_i = \sqrt{s} E_{\text{shared}} + \sqrt{1-s} \Xi_i$, where $s \in [0, 1]$ is the fraction attributable to a shared foundation model, vendor or pretraining corpus. Alignments concentrate at $r_{ij} \rightarrow s$ with $O(1/d)$ fluctuation and an explicit failure probability, so $N_{\text{eff}} \rightarrow 1 + \kappa s(N - 1)$ and every result below inherits s for free: fifty dealers fine-tuning one vendor’s model are dynamically $\sim 1 + 49\kappa$ learners at $s \approx 1$ and barely more than one at $s \approx 0$.

Why the naive diversity index fails, twice. Mean alignment understates *clustered* alignment: three tightly aligned firms among ten otherwise-orthogonal ones give $N_{\text{eff}} = 2.60$ at $\kappa = 0.8$ against 1.48 for the uniform configuration with the identical mean, so a market at $m_1 = 0.5$ runs $m_N = 1.30$ and is unstable while the mean index reports 0.74 and calls it safe with margin. The simplex is sharper: it minimizes mean alignment at $-1/(N - 1)$ yet its $\lambda_{\max} = N/(N - 1)$ exceeds orthogonality’s 1, so the mean *does not order configurations correctly* and fails as a ranking, not only as a level. The failure is signed: $N_{\text{eff}} \geq 1 + \kappa(N - 1)\bar{r}$ for every R , so the mean index always under-states risk.

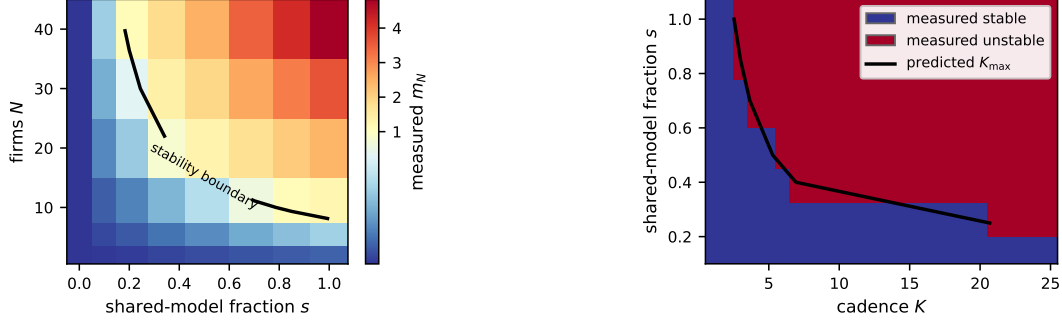


Figure 1: Left: the (N, s) phase diagram over 48 cells (Theorem 2). Right: the cadence window over 175 cells of the (N_{eff}, K) grid against predicted K_{max} , with s contours and $c = 0.8$ (Theorem 3); the realized window is $\lfloor K_{\text{max}} \rfloor$ and the plotted frontier continuous. Both are dry runs (Section 8).

5 Result 2: the crowding-cadence frontier

The first lever is quantity regulation, and it is the one a firm can pull alone: retrain less often. A learner taking only K gradient steps per deployment realizes outer-map slope $\mu(K) = -m + c^K(1 + m)$, where $c = 1 - \eta\gamma$ is the inner per-step contraction (Mendler-Dünner et al., 2020). Because c is built from the learner’s own objective curvature and step size, it does not depend on how many competitors it has, and that independence is what makes the composition below work. We verify it by measuring the realized contraction inside the joint market.

Theorem 3 (Systemic cadence window). *Under Assumption 1, in the N -firm market with synchronous K -step retraining the joint deployment map is $c^K I + (1 - c^K)J$, so the binding mode has slope $\mu_N(K) = -m_N + c^K(1 + m_N)$ and the market is stable if and only if*

$$K < K_{\text{max}} = \frac{\ln((m_N - 1)/(m_N + 1))}{\ln c}, \quad (3)$$

with K_{max} infinite when $m_N \leq 1$ and strictly decreasing in m_N .

The slope on the mode carrying $\lambda_i(R)$ is strictly decreasing in λ_i , which puts every slope at or below $c^K < 1$, so the upper side of the constraint never binds at any cadence, and places the binding mode at λ_{max} whatever the sign pattern of R .

Critical crowding. The same inequality reads $m_N < (1 + c^K)/(1 - c^K)$, and since K is a positive integer the most permissive case is $K = 1$. A stabilizing cadence exists at all if and only if $m_N < (1 + c)/(1 - c)$, a factor of 9 at $c = 0.8$. Past that the market is unstable at every retraining frequency. The laziest retrainer is therefore the hardest to destabilize.

What the supply chain does to the window. Substituting Result 1 makes K_{max} a function of vendor concentration at a fixed number of firms: at $m_1 = 0.15$, $\kappa = 0.8$, $N = 30$, $c = 0.8$, the window runs $K_{\text{max}} = 20.68$ at $s = 0.25$, 5.28 at $s = 0.5$ and 2.53 at $s = 1$. Holding competitors fixed and raising vendor concentration cuts every incumbent’s retraining budget by a factor of eight. The externality in its sharpest form is therefore not merely that **your competitor’s entry consumes your retraining budget** but that **your competitor’s choice of vendor does too**, without their entering at all.

What the lever costs. Cadence buys stability with staleness, and the exchange rate is the same quantity on both sides: after a retraining round the deployed model’s gap to its own best response is exactly c^K times what it was, so the c^K that widens the stability margin is the staleness.

6 Result 3: herd immunity, and diversity as its substitute

The second lever is a technology mandate, and unlike cadence it is not something a firm can pull alone to any effect. A *corrected* firm runs the feedback-aware update (Izzo et al., 2021; Miller et al.,

204 2021), so its retraining is governed by the objective curvature γ_{PO} in place of the cobweb and its
 205 modulus is scaled by $\theta = \gamma/\gamma_{PO} < 1$. What is new is running it inside the game: a mixed market of
 206 N_b blind firms and $N - N_b$ corrected ones, corrected fraction $\rho = 1 - N_b/N$. Correction changes a
 207 firm's gain and not its response direction, so R is untouched.

208 **Theorem 4** (Mixed-market stability). *Under Assumption 1, with supply-chain alignment and two*
 209 *correction levels, the joint Jacobian's symmetric congruence is a diagonal matrix plus a positive*
 210 *rank-one term, so its spectrum solves a secular equation. Two distinct moduli collapse it to a*
 211 *quadratic, and for $1 \leq N_b \leq N - 1$ the radius is its larger root,*

$$\rho(J) = \frac{1}{2}(P + \sqrt{P^2 - 4Q}), \quad P = (1 - \tilde{\kappa})m_1(1 + \theta) + \tilde{\kappa}m_1(N_b + \theta N_c), \quad Q = \theta m_1^2(1 - \tilde{\kappa})N_{\text{eff}}, \quad (4)$$

212 with $\tilde{\kappa} = \kappa s$ and $N_c = N - N_b$. An empty block needs the single-block form instead, the quadratic
 213 otherwise leaving a phantom root that can exceed the true radius.

214 N_{eff} appearing inside Q is why every statement below carries s : the mixed market inherits Result 1's
 215 crowding through the product term, not by stipulation.

216 **The strong-correction limit, and it errs in the unsafe direction.** As $\gamma_{PO} \rightarrow \infty$ the constant
 217 term vanishes and the unstable cycle runs through blind firms only, giving $N_b < N_c(s) = 1 +$
 218 $(1/m_1 - 1)/(\kappa s)$. That criterion taken alone is **optimistic rather than conservative**. The radius is
 219 nondecreasing in θ by Perron-Frobenius on an entrywise-nonnegative matrix, so the limit under-states
 220 it: on random draws it calls 11.8% of configurations stable that are unstable at every finite correction
 221 strength. The exact root is therefore the criterion and the limit its corollary, quoted with its error
 222 direction. The same monotonicity carries the reassuring half: **correction never backfires**, so no
 223 configuration exists in which un-blinding a firm destabilizes the market.

224 **The collapse, and its honest generalization.** At $\kappa = s = 1$ the limit reads $\rho^* = 1 - 1/m_N$,
 225 **exactly the epidemiological herd-immunity threshold** $1 - 1/R_0$ with the systemic modulus as the
 226 reproduction number. A market of 10 firms at $m_N = 2.5$ needs 60% of them un-blinded *if correction*
 227 *is perfect*. It is not, and at that same corner the quadratic degenerates and the stability condition
 228 solves exactly for

$$\rho^*(e) = \frac{1 - 1/m_N}{e}, \quad e = 1 - \gamma/\gamma_{PO}, \quad (5)$$

229 with e the **correction efficacy**. That is the standard coverage requirement for an *imperfect vaccine*,
 230 of which the clean law is the perfect-efficacy corner. The clean form assumes homogeneous mixing
 231 (Fine et al., 2011), and the collapse here assumes the $\kappa = s = 1$ corner for precisely that reason.

232 **Corollary 5** (Critical efficacy). $\rho^*(e)$ exceeds one, so no corrected fraction works at all, exactly
 233 when $\theta > 1/m_N$. Correction is a usable lever only if $\gamma_{PO}/\gamma > m_N$.

234 At $m_N = 2.5$ the corrected update must deliver a $2.5\times$ curvature improvement or un-blinding cannot
 235 stabilize the market whatever fraction is treated. This parallels Section 5's critical crowding: each
 236 lever has a regime past which it stops working, and knowing both is what makes trading them off
 237 honest rather than an extrapolation.

238 **The synthesis.** ρ^* is increasing in s , strictly wherever positive, so a market reaches stability along
 239 either axis. At $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$: a monoculture ($s = 1$) needs $\rho^* = 0.596$, that is 12
 240 corrected firms; at $s = 0.5$, $\rho^* = 0.242$, five firms; at $s = 0.2$, $\rho^* = 0$, so the market is stable with
 241 no corrected agents at all. The regulator's count is $N - \lceil N_c \rceil + 1$ rather than $\lceil \rho^* N \rceil$, and the two
 242 differ by a firm exactly when N_c is an integer.

243 **The economic reading.** Correction is a public good: an un-blinded firm captures a private benefit
 244 while the stability it contributes accrues to everyone, so the market free-rides below the threshold
 245 and does not reach ρ^* unaided (Bergstrom et al., 1986). Model diversity has the same structure, a
 246 firm choosing an idiosyncratic vendor paying privately and supplying systemic stability for free. **The**
 247 **substitution is therefore between two goods that are both under-supplied**, not between a good
 248 and a bad.

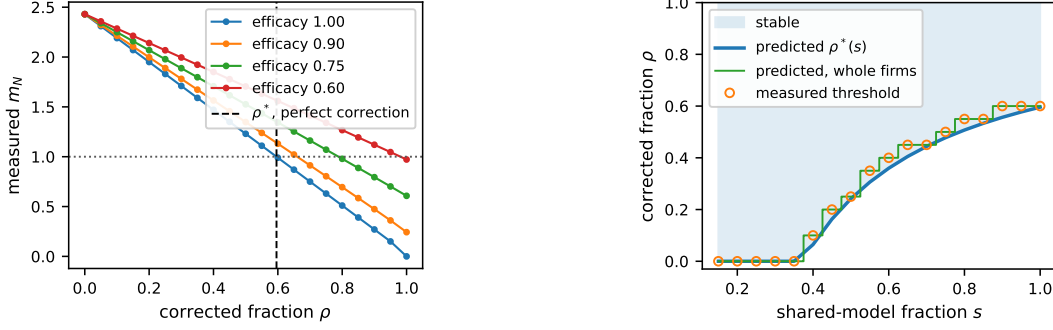


Figure 2: Left: stability against corrected fraction at $N = 20$, $m_1 = 0.15$, $\kappa = 0.8$, $s = 1$, across efficacies; the gap between the exact threshold and the perfect-correction limit is a result, not an error bar. Right: **the substitution frontier**, the (ρ, s) iso-stability curve, matching at 18 of 18 points. Both are dry runs (Section 8).

7 Result 4: the Pigouvian wedge

The third lever is the price instrument. Inside the stable region the binding mode is an AR(1) with coefficient $-m_N$ driven by noise of variance σ^2 , so its stationary variance is $V(m_N) = \sigma^2/(1-m_N^2)$: the smooth proxy for divergence risk, divergent exactly at the boundary the rest of the paper is about, and nothing below uses more than V being positive, increasing and convex. The sign the variance does not see is not lost: the lag-1 autocorrelation is $-m_N$ and therefore *negative*, so a crowded market's common mode oscillates rather than persists, which is the observable the supervision estimator of Section 9 reads.

Each firm picks an adaptation aggressiveness a_i , meaning cadence, learning rate or responsiveness, buying concave tracking benefit $B(a_i)$ through $m_i = \mu(a_i)$. Firm i bears weight w_i on common-mode variance and the planner W , with $W > \sum_i w_i$ because every trade has a client bearing execution-quality variance that no dealer's objective contains. That is an assumption, and the results turn on its *sign* and never its size.

Lemma 6 (Marginal crowding share). *At the symmetric point, $\partial m_N / \partial m_i = N_{\text{eff}} v_i^2$ where v is the leading eigenvector of the coupling matrix. The shares sum to N_{eff} rather than to one, and on exchangeable R each equals N_{eff}/N .*

Where a sign condition is needed, and where it is not. Theorem 2's radius needs none: R is positive semidefinite with unit diagonal, so every $\lambda_i(R) \geq 0$ and $1 + \kappa(\lambda_i - 1)$ increases in λ_i , placing the radius at $\lambda_{\max}$ whatever the off-diagonal signs. What needs one is the *reading* of Lemma 6, that the binding mode is the common one so v is a vector of market shares. (A5) supplies it: with R entrywise nonnegative, Perron-Frobenius places the leading eigenvector in the nonnegative orthant, and shared vendors and corpora produce positive alignment.

Both readings of Lemma 6 matter. The *sum* is N_{eff} , so the market's total sensitivity to a uniform increase in aggressiveness is amplified by exactly the effective learner count of Result 1. The *individual* share is N_{eff}/N , the arithmetic shape of every commons problem. The naive guess here is $1/N$ by analogy with the commons, and it is wrong by the factor N_{eff} , which at $N = 20$, $\kappa = 0.8$, $s = 1$ is 16.2.

Theorem 7 (The wedge). *Under Assumption 1 and the welfare assumptions above, at a symmetric profile the private and social first-order conditions are the same equation with a different price on crowding, and the marginal cost a firm ignores is*

$$t^* = (W - w_i) V'(m_N) \frac{N_{\text{eff}}}{N} \mu'(a), \quad V'(m_N) = \frac{2\sigma^2 m_N}{(1 - m_N^2)^2}. \quad (6)$$

A per-unit fee t^* on aggressiveness makes the private condition coincide with the social one, so the taxed decentralized equilibrium implements the symmetric social optimum.

With symmetric weights the private condition ignores the fraction $(N-1)/N$ of the marginal variance cost borne by other firms, plus the entire client-side exposure. t^* is strictly increasing in N , κ , s and

Table 2: Panel coverage and status. Panel 1 is external validation against a prior published run, not a self-consistency check.

Panel	Tests	Status	Outcome
1. Amplification	Section 3	measured	$1.7428\times, 3.1567\times$; relative error 0.00
2. (N, s) phase diagram	Thm 2	dry run	48 cells, max error 7.1×10^{-15}
2b. Clustered companion	Thm 2	dry run	measured m_N 1.30; mean index says 0.74
3. Cadence frontier	Thm 3	dry run	175 cells, zero disagreements
4. Herd immunity	Thm 4	dry run	thresholds 12/14/16/20 firms across efficacy
5. Substitution frontier	Thm 4	dry run	18 of 18 points exact
6. Over-adaptation	Thm 7	dry run	12 configurations, zero contradicting Cor. 8

m_N , and diverges at the boundary like $(1 - m_N)^{-2}$. The rate is the economically loaded part: a regulator setting a fee from a comfortable-looking market and holding it fixed is setting it far too low by the time the market is close to instability.

The provenance channel. m_N depends on s , so the wedge has a second channel: $\partial m_N / \partial a_i = (N_{\text{eff}}/N)\mu'(a_i)$ against $\partial m_N / \partial s = m_1\kappa(N - 1)$. The same wedge therefore prices **shared-model adoption** directly, turning the monoculture externality into a fee rather than a warning, and the two channels are asymmetric. The aggressiveness factor N_{eff}/N is bounded and tends to κ from above, so a firm's marginal footprint through its own cadence does not grow without bound, while the provenance channel is **linear in N and does not decay** against a private adoption gain that does not grow with market size at all. Unregulated adoption therefore runs toward the unstable configuration, which is why the diversity floor of Section 6 is a different instrument from the cadence cap of Section 5 rather than a restatement of it.

Corollary 8 (Over-adaptation). *The decentralized equilibrium over-adapts relative to the social optimum for every $N \geq 2$, strictly so even with client exposure switched off, and the distortion grows as the market crowds. At $N = 1$ with no client exposure the wedge is exactly zero.*

The fraction the private condition ignores is free of N_{eff} : with no client exposure it is exactly $(N - 1)/N$ whatever κ and s are. Crowding enters the wedge's *level*, through $V'(m_N)$, and not the distortion's *shape*, so Corollary 8 is the generic commons result and the provenance asymmetry above is the market-of-models one. Sections 5, 6 and 7 are respectively quantity regulation, a technology mandate or structural remedy, and the price instrument; choosing among them under uncertainty is Weitzman's question (Weitzman, 1974), and we supply the objects needed to pose it rather than answering it.

8 Experiments

Six panels, one per result, each with its evidential status in its own row of Table 2, because the six are not the same kind of evidence. A panel marked **measured** ran in the order-flow simulator, a genuine N -dealer market with informed flow, spread and inventory that reduces bit for bit to the single-dealer market at $N = 1$. A panel marked **dry run** ran in a linearized reference environment realizing the model's response geometry and nothing else: it checks that the closed forms were derived from the map we believe they were derived from, which is a check on the derivation rather than evidence about dynamics, and it has caught real errors. It establishes nothing about a market. Stability everywhere is estimated from the trajectory's asymptotic growth rate under the actual retraining map rather than read off an eigensolve. All runs are CPU-only and deterministic from (config, seed), with common random numbers on paired probes; every closed form carries a numerical certificate and the harness exits nonzero on any disagreement. The simulator is described generically pending an artifact release; code, derivations and certificates are available to reviewers as anonymized supplementary material.

The one measured panel. Panel 1 reproduces a prior published run in that run's own simulator, at the relative error Table 2 records. **The gap to the linear prediction is content, not error:** the market runs -12.9% at $N = 2$ and $+5.2\%$ at $N = 3$, with *opposite signs*, which is what a nonlinear market with saturating flow does around a linearization. The reference environment reproduces the prediction exactly because it omits precisely that nonlinearity. The differential mode measures

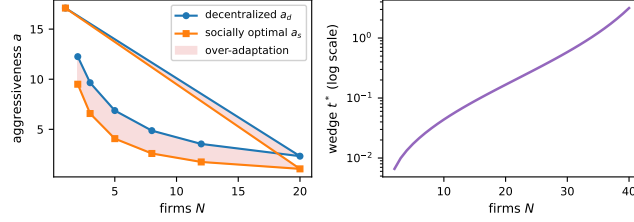


Figure 3: Over-adaptation and the wedge. Left: decentralized against socially optimal aggressiveness across N at $\chi = 1$, $s = 1$. Right: t^* against N , log scale. Dry run; the boundary divergence fits $(1 - m_N)^{-2}$ at log-log slope -2.000 .

323 3.4×10^{-3} against a theoretical zero at $\kappa = 1$, so the instability is purely common-mode, the
 324 mechanism Theorem 2 generalizes.

325 **Why the rest are not measured.** A heterogeneous-response port of the order-flow simulator was
 326 built and is exact at flat exposure profiles, reproducing the unmodified simulator at every N from 1 to
 327 6, but its acceptance gate failed: the liquidity and price-impact channels stay shared whatever the
 328 response profiles are, a residual reaching 17.7 percentage points at $N = 2$, larger than several effects
 329 these panels resolve, and at $N \geq 4$ the finite-difference probe no longer measures a local slope. We
 330 drew no figure from that port and moved no status. Panel 6 is a dry run for a different reason: the
 331 order-flow simulator carries no aggressiveness choice variable and no welfare object, so no measured
 332 counterpart exists to be built.

333 In panel 6 both first-order conditions are solved by bisection with m_N read from realized dynamics
 334 rather than the closed form, so the over-adaptation verdict is a property of the simulated market rather
 335 than of the algebra: the gap widens from 29.0% at $N = 2$ to 119.9% at $N = 20$, and the degenerate
 336 $N = 1$ case sits at exactly zero, as Corollary 8 requires.

337 9 Limitations and conclusion

338 **Supervision from public prices.** Every quantity above is hidden from the regulator these results
 339 address, and near the boundary the system tells on itself: as m_N approaches one, the leading principal
 340 component of public quotes has lag-1 autocorrelation approaching one in magnitude, negative because
 341 the common mode oscillates, and variance share growing like $1/(1 - m_N)$. A supervisor can therefore
 342 estimate distance to instability from public prices alone, and the estimate sharpens as the market
 343 nears the boundary it watches for. Its consistency proof and detection sample complexity are deferred,
 344 and the appendix's real-data panel is consistency evidence with a placebo, never identification.

345 **Limitations.** The analysis is linearized around the joint equilibrium (A1), and panel 1 shows what
 346 that costs in a real market. **The heterogeneous sweep is closed-form agreement, not measurement,**
 347 the largest gap in the work; routing the liquidity and impact channels through the response profiles
 348 is the identified repair. Theorem 2 is exact for equal moduli (A2), unequal ones giving a two-sided
 349 bound, and share-weighted alignment, the block-secular reduction when (H1) fails, and the sharp
 350 concentration rate are named extensions. The $1 - 1/R_0$ collapse assumes the $\kappa = s = 1$ corner,
 351 three or more correction levels return a genuine secular equation, and correction moving a response
 352 direction rather than a gain is not modelled. Deployment is synchronous (A3); heterogeneous clocks
 353 break the eigenvector sharing Theorem 3 uses. Theorem 7 prices instability through stationary
 354 variance, a modeling choice. Neither R nor s is observable to an outsider, which the estimator above
 355 answers with its real-data leg deferred.

356 **Conclusion. Individual model evaluation cannot certify a market of models.** The quantity
 357 separating the two questions is the effective number of independent learners, which no evaluation
 358 of a single model can see: a firm can pass every stability test available to it and still be one of the
 359 reasons its market fails. Regulation of shared general-purpose models has thresholds on model scale
 360 but no mechanism tying model sharing to a measurable systemic quantity. This paper supplies one,
 361 with two substitutable instruments and a price.

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A Technical appendices

Complete proofs of every numbered result in the body, the numerical certificates and their tolerances, the full experimental specifications, and the supervision panel with its placebo are provided in the supplementary material. Every closed form in the body is checked by an assertion-based certificate that fails loudly; 525 assertions across seven files pass at submission time.

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